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SPECIAL ARTICLE

# Clinical suspicion, diagnosis and management of cardiac amyloidosis: update document and executive summary

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*Abbreviations:* 99mTc-DPD, Technetium-99m-labeled 3,3-diphosphono-1,2-propanodicarboxylic acid; 99mTc-HMDP, Technetium-99m-labeled hydroxymethylene diphosphonate; 99mTC-PYP, Technetium-99m-labeled pyrophosphate; CA, Cardiac amyloidosis; AL, Amyloid light chain amyloidosis (primary amyloidosis); AL-CM, Light chain amyloidosis cardiomyopathy; ATTR, Transthyretin amyloidosis; ATTR-CM, Transthyretin amyloid cardiomyopathy; ATTRv, Variant transthyretin amyloidosis (hereditary); ATTRwt, Wild-type transthyretin amyloidosis (spontaneous); AVB, Atrioventricular block; Cardiac MRI, Cardiac magnetic resonance imaging; CM, Cardiomyopathy; AoS, Aortic stenosis; ECG, Electrocardiogram; AF, Atrial fibrillation; FAQs, Frequently asked questions; GDS–FAST, Global Deterioration Scale (GDS), Functional Assessment Staging (FAST); MGUS, Monoclonal gammopathy of uncertain significance; HF, Heart failure; HFpEF, Heart failure with preserved ejection fraction; ICyFA, Heart Failure and Atrial Fibrillation Working Group; PM, Pacemaker; NT-proBNP, N-terminal-pro-brain natriuretic peptide; PADO, Predicted age of disease onset; RAISE score, Remodeling, Age, Injury, Systemic and Electrical disorders score; MRI, Magnetic resonance imaging; SARC-F, Strength, Assistance with walking, Rising from a chair, Climbing stairs, and Falls questionnaire; SEMI, Spanish Society of Internal Medicine; SPECT, Single Photon Emission Computed Tomography; CTS, Carpal tunnel syndrome; TAVI, Transcatheter aortic valve implantation; TTR, Transthyretin; ECV, Extracellular volume; CGA, Comprehensive geriatric assessment.

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**KEYWORDS**
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*Abstract:* In recent years, the interest in cardiac amyloidosis has grown exponentially. However, there is a need to improve our understanding of amyloidosis in order to optimise early detection systems. Therefore, it is crucial to incorporate solutions to improve the suspicion, diagnosis and follow-up of cardiac amyloidosis. In this sense, we designed a tool following the different phases to reach the diagnosis of cardiac amyloidosis, as well as an optimal follow-up: a) clinical suspicion, where the importance of the ''red flags'' to suspect it and activate the diagnostic process is highlighted; 2) diagnosis, where the diagnostic algorithm is mainly outlined; and 3) follow-up of confirmed patients. This is a practical resource that will be of great use to all professionals caring for patients with suspected or confirmed cardiac amyloidosis, to improve its early detection, as well as to optimise its accurate diagnosis and optimal follow-up.


**PALABRAS CLAVE**
Amiloidosis cardíaca;
Insuficiencia cardíaca;
*Red flags*;
Sospecha clínica;
Diagnóstico precoz

**Sospecha clínica, diagnóstico y seguimiento de la amiloidosis cardíaca: documento de actualización y resumen ejecutivo**

**Resumen** En los últimos años el interés en la amiloidosis cardíaca ha crecido exponencialmente. No obstante, es necesario mejorar en su conocimiento para optimizar los sistemas de detección precoz. Por lo tanto, es crucial incorporar instrumentos para mejorar la sospecha, diagnóstico y seguimiento de la amiloidosis cardíaca. En este sentido, diseñamos una herramienta siguiendo las distintas fases para alcanzar el diagnóstico de la amiloidosis cardíaca, así como un óptimo seguimiento: a) sospecha clínica, donde se pone en relieve la importancia de los ''*red flags*'' para sospecharla y activar el proceso diagnóstico; 2) diagnóstico, donde se desgrana principalmente el algoritmo diagnóstico; y 3) seguimiento de los pacientes confirmados. Se trata de una herramienta que será de gran utilidad para todos aquellos profesionales que atienden a pacientes con sospecha de amiloidosis cardíaca o enfermedad ya confirmada, para mejorar su detección precoz, así como optimizar su preciso diagnóstico y un óptimo seguimiento.


## Introduction

Despite the optimization of pharmacological and nonpharmacological tools,[1,2] heart failure (HF) continues to be a global public health problem due to its high prevalence and incidence[3,4] as well as its significant morbidity and mortality.[5,6] It is the main cause of hospitalization of patients over 65 years of age in internal medicine departments in our setting.[7] In recent years, cardiac amyloidosis (CA) has emerged as a relevant etiology of HF that should not be underestimated.[8,9] Since the standardization of noninvasive diagnostics[10] and the possibility of prescribing specific treatment,[11] interest in CA has increased significantly.[12]

Of the different causes of CA, most are due to the two subtypes of transthyretin amyloidosis (ATTR) defined according to the presence or absence of mutation in the gene that encodes transthyretin (TTR): wild-type (ATTRwt), with no mutation identified in the TTR gene—probably the most frequent CA—and the hereditary or variant form (ATTRv),[13] even ahead of the more classically recognized light chain amyloidosis (AL).[14] The prevalence of ATTR in HF remains uncertain,[15–18] but is estimated to be around 13% in patients with HF with preserved ejection fraction (HFpEV).[16] Recent studies have indicated that this prevalence is much higher (over 20%) if patients with LVEF < 50% are also included.[17,18]

Therefore, this is a subgroup of patients with HF that must be identified because of the potential clinical and financial impact they may have on Spain's healthcare system in the coming years.[19] Moreover, patients with ATTR cardiomyopathy (CM) (ATTR-CM) may have disease that progresses silently or mimics other conditions and the diagnosis must be made in these early stages.[20] This is why a high degree of suspicion is essential for an early diagnosis.[21]

Nevertheless, it continues to be necessary to improve knowledge of the disease in order to optimize early detection systems and diagnostic procedures.[17] In this regard, it is crucial to provide healthcare professionals with tools to improve CA diagnosis and follow-up. For this reason, the Heart Failure and Atrial Fibrillation Working Group (ICyFA, for its initials in Spanish) of the Spanish Society of Internal Medicine (SEMI, for its initials in Spanish) has designed a pocket tool to improve the clinical suspicion, diagnosis, and care of patients with CA.[22]

## Methodology

Given the clinical relevance of—especially ATTR-CM—in the present and future of HF, a brief tool has been designed for daily use by professionals who care for patients with HF in order to improve the suspicion, diagnosis, and management of patients with CA.[21–25]

This document focuses on the different phases of diagnosing CA as well as optimal follow-up: a) clinical suspicion; 2) diagnostic process; and 3) follow-up on patients once the diagnosis is confirmed. It also includes a proposal of a multidimensional assessment to evaluate the suitability of indicating specific treatments (Additional material - Appendix S1).

## Clinical suspicion

An early diagnosis of CA is key given that patients' vital prognosis progressively declines due to continuous amyloid deposition.[14,26] Moreover, an accurate diagnosis is crucial given that CA entails a differential therapeutic approach and a worse prognosis compared to other HF etiologies.[27] In this regard, although survival in ATTRwt-CM is longer than that of AL-CM, the natural course of ATTRwt-CM is less favorable than that of other cardiomyopathies.[28] Therefore, suspecting the disease is imperative in order to change the disease course.[29]

Amyloid fiber accumulation predominantly in the septal-basal area will cause interventricular septum thickening and left ventricle stiffening that can lead to diastolic dysfunction.[30] When this is accompanied by a combination of clinical features, there may be a high level of suspicion and the diagnostic process can start. Therefore, experts suggest that septal thickening ($\geq$12 mm) together with the presence of one or more ''red flags'' may mean a highly likely diagnosis of ATTR-CM and thus the patient should be screened for the disease (Fig. 1).[22,23]

### «Red flag» #1: clinical manifestations

CA can manifest as a wide range of signs and symptoms (''red flags'') which, together with compatible imaging tests, are helpful in suspecting the disease.[21–23] The most common clinical manifestation of CA is HF (mainly in patients >65 years).[8] This is the most common form of presentation in patients with ATTRwt (between 50% and 85% of cases). In ATTRv, cardiac involvement will depend on the expressed mutation of the TTR gene.[16] Amyloid fibers can infiltrate any cardiac structure, giving rise to different clinical manifestations depending on the location and degree of deposition of amyloid material.[8]

When the accumulation of amyloid fibers is predominantly in the valvular area, it will give rise to heart valve diseases, with low-flow, low-gradient aortic stenosis being very characteristic, particularly in patients >65 years.[31] A prevalence of ATTRwt that ranges from 5% to 16% has been described in patients with AoS.[32] In addition, its concomitant diagnosis with CA entails a worse prognosis compared to AoS alone unless specific treatment is given.[33] In this regard, treatment with transcatheter aortic valve implantation (TAVI) has proven to be very useful.[34] To aid in the early diagnosis of CA in patients with AoS, a scoring system (RAISE score) has recently been designed to identify patients with a higher probability of CA. CA screening is recommended in patients with a RAISE score $\geq$2, given its high diagnostic probability (Additional material - Appendix S2).[35]

On an extracardiac level, there has been growing interest in the study of the musculoskeletal manifestations related to CA in recent years.[36] This is because they can be highly significant both for diagnosis, as it can precede other manifestations by up to 15 years, and for prognosis.[37] Furthermore, it is estimated that they may be present in up to 80% of patients with ATTR-CM and are more characteristic in those with ATTRwt.[38] Among these, the most frequent and well known is CTS, which may be present in 33% to 64% of patients with ATTRwt and around 60% of those with ATTRv.[39–41] However, other manifestations must be considered, such as lumbar canal stenosis (which may be present in up to 64% of patients with ATTRwt)[42] or a history of knee, hip, and/or shoulder arthroplasty, present in nearly 50% of these patients depending on the series,[43] among others.[44,45] Performing a biopsy during surgery could be very useful for the early diagnosis of ATTR in some cases.[46]

### «Red flag» #2: electrocardiographic findings

The most frequent electrocardiographic (ECG) finding in CA (up to 70% depending on the series) is a pseudo-infarction pattern in the anteroseptal area.[14] Also, although it is not the most frequent, one of the classic patterns is discordance between left ventricular wall thickness and QRS voltages. In CA, the increase in myocardial thickness is a pseudohypertrophy due to myocardial infiltration. This is what leads to the characteristic discrepancy between normal or low QRS voltages and increased myocardial thickness.[29,47] It is present in no more than 20% of patients with ATTR-CM and is more prevalent in AL-CM (60%).[14] Also noteworthy is the presence of both atrial electrical conduction disorders (mainly atrial fibrillation (AF)) in 43% to 67% of patients with

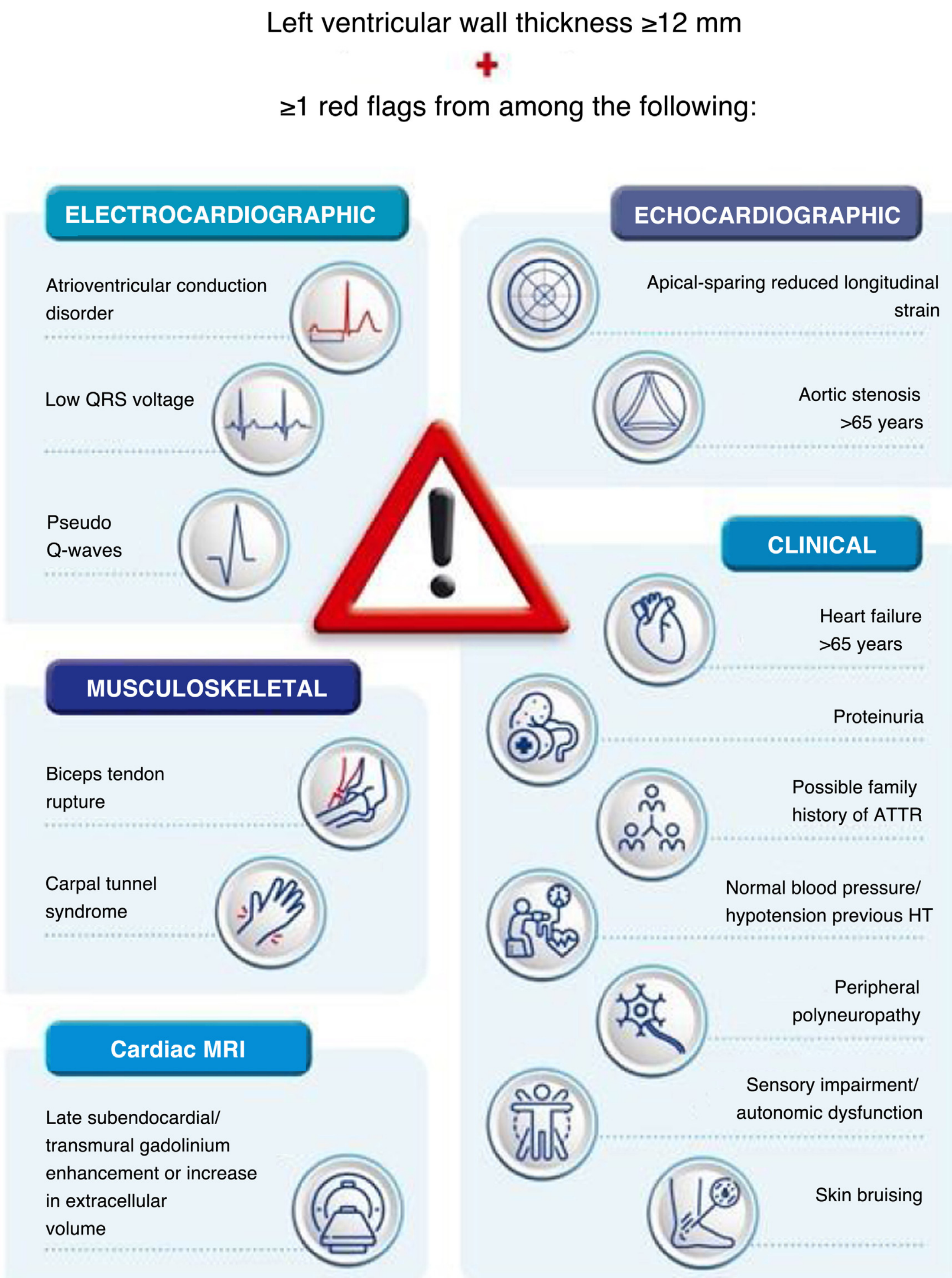



**Figure 1** ''*Red flags*'' for the clinical suspicion of cardiac amyloidosis.
ATTR: Transthyretin amyloidosis; HT: hypertension.

ATTRwt[16,47,48] and intraventricular conduction disorders and atrioventricular block (AVB) in up to 22% of cases.[30] This is due to infiltration of the conduction system by amyloid fibers. It can affect any area of the conduction tissue and sometimes requires pacemaker (PM) implantation (Supplementary material - Appendix S3).

## «Red flag» #3: echocardiographic findings

Classically, CA has been identified with left ventricular hypertrophy that is predominantly septal.[49] In fact, the presence of septal thickening ≥12 mm is an essential condition for starting the study of a possible case of CA.[22,23] It is usually associated with concentric, biventricular hypertrophy, but up to 23% of patients have asymmetric hypertrophy,[48] which is why CA is one of the most frequent phenocopies of hypertrophic cardiomyopathy[50] (Additional material - Appendix S4).

CA has also classically tended to be associated with preserved or slightly decreased LVEF, but recent series suggest that the disease can present with any degree of LVEF.[51] Notably, although LVEF is preserved in most cases of CA, these patients have impaired myocardial function which entails reduced systolic volume. This feature will be more relevant in cases in which CA and AoS coexist, which will be characterized as being low flow-low gradient due to this decrease in systolic volume.[30,34]

One of the echocardiographic abnormalities that can most help in differentiating CA from other cardiomyopathies is strain. In CA, there is typically a decrease in longitudinal strain in the basal and middle segments with preservation of the apical segments, referred to as the ''cherry on top'' or ''Japanese flag'' pattern[21,47,52] (Supplementary material - Appendix S5).

In recent years, an echocardiographic score has been proposed to facilitate the diagnosis of AC in the presence of increased left ventricular wall thickness (Additional material - Appendix S6).[23,53]

### «Red flag» #3: cardiac MRI findings

In recent years, cardiac MRI has revolutionized the noninvasive diagnosis of cardiomyopathies because it lends greater precision to the functional and structural assessment of the heart than what can be obtained with echocardiography as well as better tissue characterization through gadolinium administration. Therefore, it has become a very important tool in the early detection of CA and in its differential diagnosis.[54]

There are three key elements in the study of CA using cardiac MRI: a) the presence of late enhancement (high sensitivity and specificity (85%–90%) for the diagnosis of CA): a global subendocardial pattern (25% of cases) is highly suggestive of CA[8,14]; b) elevated native T1 values using the T1 mapping technique[30]; and c) calculation of extracellular volume (ECV) after contrast administration: myocardial amyloid fiber deposits increase ECV and result in the accumulation of gadolinium contrast identified with cardiac MRI.[55] Since these are quantitative parameters, both ECV calculation and T1 mapping techniques can be very useful not only for diagnosis but also for follow-up, as well as for evaluating treatment response (Supplementary material - Appendix S7).[56]

Similar to the criteria previously described in echocardiography, cardiac MRI criteria have been proposed that would allow for optimizing the diagnosis of CA (Additional material - Appendix S8).[23]

## Diagnostic algorithm

The diagnostic process for CA is based on the combination of hematologic tests and cardiac scintigraphy (Fig. 2).[10,21–23]

### Hematological study

A complete hematological study should be performed in all patients with suspected CA by analyzing the monoclonal component (immunofixation) in blood and 24-h urine as well as the analysis of serum free light chains, with the aim of ruling out blood dyscrasias.[10] Their combined study is essential for ruling out the presence of AL or other significant hematologic involvement if the study results are strictly normal.[57] In cases in which the hematologic study shows any abnormality, a specific study and/or evaluation by the hematology department will be required. Up to 5% of the population older than 65 years has monoclonal gammopathy of uncertain significance (MGUS)[29] and in older adults, a moderate increase in circulating light chains should not lead directly to an AL diagnosis.[8] The kappa and lambda light chain ratio should be used for diagnostic and disease monitoring purposes. In regard to its interpretation, the ratio is more useful in the diagnosis than absolute values of kappa and lambda chains, especially in the presence of renal disease, as the clearance of kappa chains is much more efficient than that of lambda chains, such that a low kappa/lambda ratio is always pathological. For normal renal function values, the normal ratio will oscillate between 0.26 and 1.65 and in the most extreme cases—patients with a glomerular filtration rate below 30 ml/min/1.73 $m^2$—a ratio of 0.54–3.30 can be considered normal.[58] Therefore, a correct interpretation of the results is essential for an optimal diagnosis and collaboration from the hematology department will be required in some cases, as discussed above.

### Cardiac scintigraphy with SPECT

The incorporation of scintigraphy in the diagnosis of CA has truly been revolutionary. It has proven to be an effective tool for locating cardiac ATTR deposits. Diphosphate (99mTc-DPD), pyrophosphate (99mTc-PYP), and hydroxymethylene diphosphonate (99mTc-HMDP)—all labeled with 99mTc (technetium)—have shown high sensitivity and specificity for ATTR fiber uptake.[10]

The scintigraphy study is interpreted based on the result of the Perugini visual score, which compares the degree of radiotracer uptake in the bone with cardiac uptake.[59] Perugini et al. observed cardiac uptake in 15 patients with ATTR and its absence in 10 with LA, defining the different degrees of visual uptake as follows: Perugini Score of 0 (Grade 0), no cardiac uptake and normal bone uptake; Perugini Score of 1 (Grade 1), mild cardiac uptake, less than bone uptake; Perugini Score of 2 (Grade 2), moderate cardiac uptake accompanied by attenuated bone uptake (bone-like uptake); Perugini Score of 3 (Grade 3), high cardiac uptake with mild/absent bone uptake (more uptake than bone) (Fig. 3).[59] Several studies have reported similar results since this definition,[60] but more recently, Gilmore et al.[10] were able to demonstrate the possibility of making a noninvasive diagnosis of ATTR by combining the scintigraphy study with a complete hematologic study. Uptake grades 2 or 3 on the scintigraphy with a normal hematologic study provided a specificity and positive predictive value for ATTR-CM of 100% (confidence interval of 98–100).[10,23]

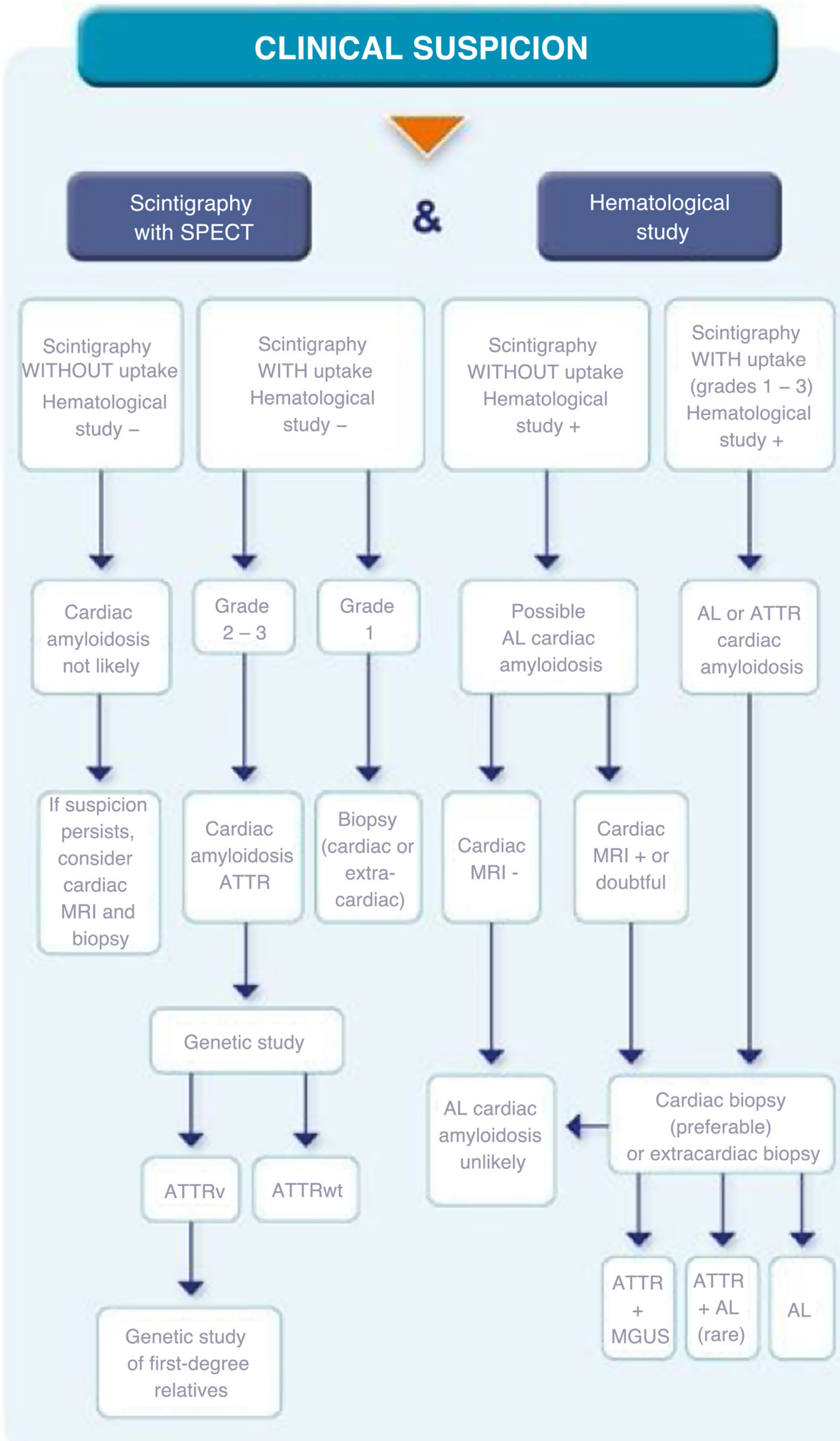



**Figure 2** Diagnostic algorithm in patients with clinical suspicion of cardiac amyloidosis.
AL: light chain amyloidosis; ATTR: transthyretin amyloidosis; ATTRv: variant transthyretin amyloidosis; ATTRwt: wild-type transthyretin amyloidosis; MGUS: monoclonal gammopathy of uncertain significance; MRI: magnetic resonance imaging; SPECT: single-photon emission computed tomography.

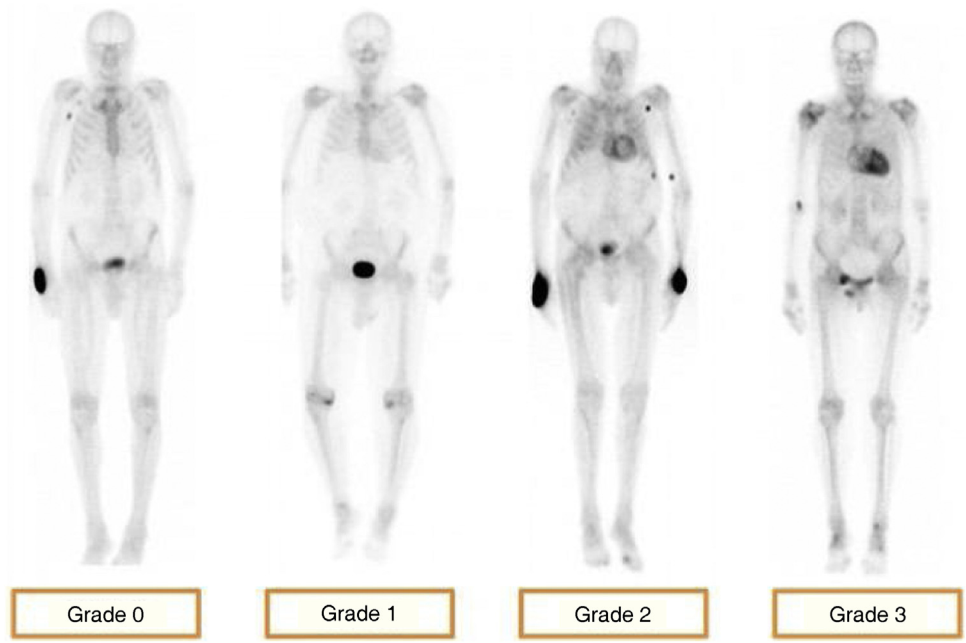



**Figure 3** Visual Perugini grading score based on biventricular cardiac uptake compared to bone.

In recent years, there has been growing interest and a need for the integration of SPECT, together with cardiac scintigraphy, in the CA diagnostic process.[61,62] Its incorporation was motivated by the fact that in different clinical contexts, there may be false negatives (in cases of very mild involvement or certain TTR mutations such as Val30Met, among others) and false positives (up to 30% of patients with LA may present myocardial radiotracer uptake or other less frequent causes of CA or recent myocardial infarction, among others) in scintigraphic uptake.[23] A SPECT study can improve diagnosis in these situations, which should always be considered when interpreting the scintigraphy results.

## Biopsy

Historically, the definitive diagnosis of amyloidosis has been made through Congo red staining tissue biopsy, which shows the pathognomonic apple green birefringence of amyloid deposits when observed under polarized light.[8,14] However, while this remains the gold standard for diagnosis of the disease, its use has now been relegated to cases of high clinical suspicion with equivocal or non-confirmatory results on noninvasive studies (Supplementary material - Appendix S9).[22,23,29]

## Genetic study

Since it is not possible to distinguish between the different ATTR subtypes either clinically or histologically, once the diagnosis is confirmed, it is essential to complete the study with complete sequencing of the TTR gene in order to establish a definitive diagnosis.[21,23] In cases in which a mutation is identified, this will allow for offering genetic counseling to patients and relatives as well as monitoring and follow-up for asymptomatic carrier relatives.[8] Recommendations from experts indicate that in asymptomatic patients who are carriers of genetic mutations of TTR variants, assessment and follow-up should begin ten years before the symptoms onset took place in the index case. The assessment should include a targeted medical history, a comprehensive physical examination (cardiological, neurological, and/or ophthalmological), as well as periodic additional examinations (ECG, echocardiogram, scintigraphy, among others) and follow-up over time.[63]

## Interpretation of results

Based on the results of the scintigraphy with SPECT and the hematological study, there are four clinical scenarios that must be interpreted in order to make an accurate diagnosis[23]:

1 Scintigraphy WITHOUT uptake and a NEGATIVE hematological study: the probability of CA will be very low, but, depending on the degree of diagnostic suspicion, completing the study with a cardiac MRI and/or, in cases with a high degree of suspicion, endomyocardial biopsy can be considered.
2 Scintigraphy WITH uptake and a NEGATIVE hematologic study: in this situation, the definitive diagnosis will depend on the degree of scintigraphic uptake. In the presence of mild uptake (Grade 1), an ATTR diagnosis cannot be confirmed and a cardiac or extracardiac biopsy is necessary to complete the study. On the other hand, higher degrees of uptake (Grades 2–3) in the absence of blood dyscrasia allows for confirming an ATTR diagnosis without the need for a biopsy.
3 Scintigraphy WITHOUT uptake and a POSITIVE hematological study: the suspicion of AL is high and it will be

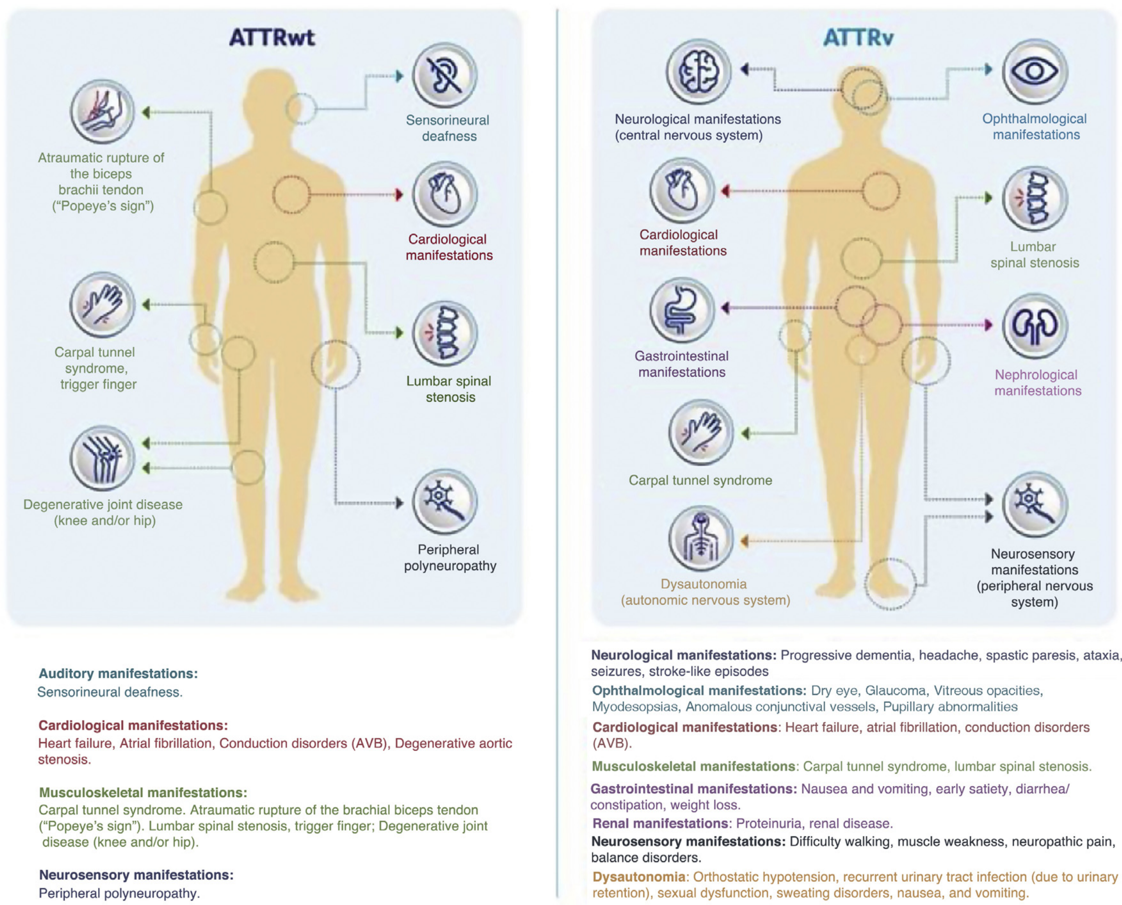



**Figure 4** Clinical manifestations of the two transthyretin amyloidosis subtypes: wild-type and variant. ATTRv: variant transthyretin amyloidosis; ATTRwt: wild-type transthyretin amyloidosis; AVB: atrioventricular block.

imperative to complete the study with a cardiac or extracardiac biopsy to demonstrate or rule out amyloid deposits.

4 Scintigraphy WITH uptake and a POSITIVE hematological study: an etiological diagnosis cannot be established with certainty with these findings (it may be AL; an ATTR with some type of hematological disease, such as MGUS; or the coexistence of ATTR and AL). Therefore, a biopsy (endomyocardial, preferably, or extracardiac, depending on each case, or even both) will be essential for determining the amyloid subtype that causes the cardiac disease.

## Clinical manifestations

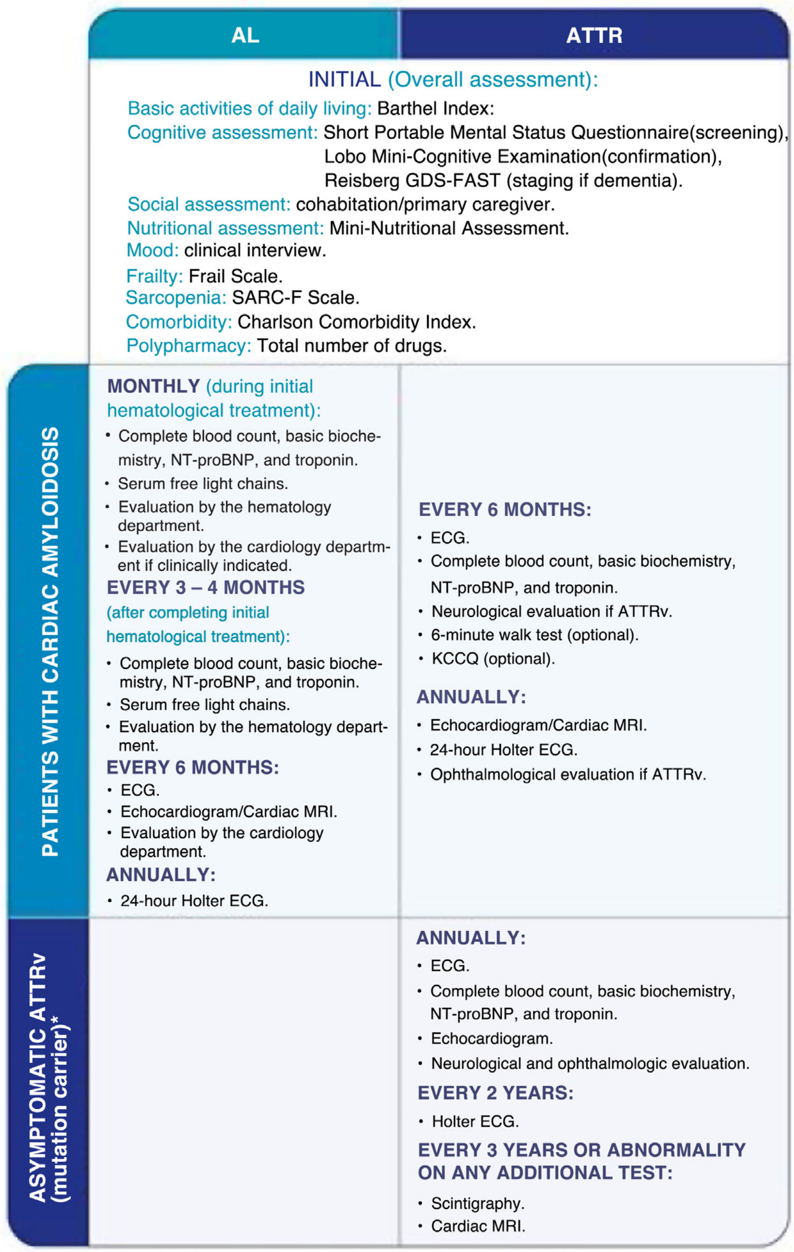
Although ATTR-CM commonly presents only with cardiological symptoms, amyloidosis is a systemic disease and thus there may be various noncardiological signs and symptoms.[8] Ophthalmological, neurological, or gastrointestinal symptoms are all noncardiological symptoms that can occur in patients with ATTR, particularly in patients with ATTRv.[24,64–66] This is why clinicians must know the clinical particularities of the two types of ATTR (ATTRv and ATTRwt) to improve their diagnosis (Fig. 4).

Clinically, ATTRwt, unlike ATTRv, presents with constant cardiac involvement, given the cardiac tropism of TTR and the predominant deposition in this location, with extracardiac manifestations limited to musculoskeletal manifestations.[8,9,36,48,49] The most prevalent extracardiac manifestation in ATTRwt is the presence of CTS, whose onset is usually precedes cardiac involvement by several years (five to ten years before cardiomyopathy).[39–41] Regarding extracardiac, nonmusculoskeletal manifestations, the presence of peripheral polyneuropathy (in about 10% of cases) or sensorineural hearing loss can be observed in patients with ATTRwt, with a clearly residual prevalence and clinical impact in these patients.[67,68]

In ATTRv, symptoms depend on the mutation and environmental factors.[29] Although cardiac and neurological manifestations are more common and better known, unlike ATTRwt, it should be considered a systemic disease that can have multiple clinical manifestations: dysautonomia, given the involvement of the autonomic system, with ophthalmologic, renal, neurological involvement; or the presence of CTS but, as stated above, with a lower prevalence than in patients with ATTRwt and, as with the rest of the clinical manifestations, it will depend on the mutation demonstrated in the TTR gene.[65–68]

## Follow-up

Once the diagnosis of CA has been confirmed, an assessment and follow-up plan must be established and coordinated in order to offer monitoring and treatment strategies




*Follow-up should be started ten years before the predicted age of disease onset (PADO). The PADO is calculated based on the mutation, the typical age of onset for that mutation, and the earliest age of onset in the family. However, if the patient wishes, a complete baseline visit can be performed at the time of genetic diagnosis followed by periodic follow-up. Table adapted from García-Povía P. et al. Eur Heart J 2021.

**Figure 5** Global assessment and follow-up on patients with a confirmed diagnosis of cardiac amyloidosis.
AL: light chain amyloidosis; ATTR: transthyretin amyloidosis; cardiac MRI: cardiac magnetic resonance imaging; ECG: electrocardiogram; GDS-FAST: Global Deterioration Scale (GDS), Functional Assessment Staging (FAST); KCCQ: Kansas City Cardiomyopathy Questionnaire; NT-proBNP: N-terminal-pro-Brain Natriuretic Peptide; SARC-F: Strength, Assistance with walking, Rising from a chair, Climbing stairs, and Falls questionnaire.

for complications and the disease itself.[23] In this regard, it must not be forgotten that in older adult patients with CA, a holistic assessment (comprehensive geriatric assessment - CGA) will be key in order to select patients who are candidates for specific treatment for the disease.[69]

It is essential to design multidisciplinary care models for CA that are patient-centered and coordinated by the nursing department. The aim of these models is to offer strategies for education, empowerment, self-management, and treatment of the disease to improve its prognosis.[70,71] In this sense, the internal medicine department, due to its own idiosyncrasy, can play a significant role in the coordination of these CA care units.

Once an optimal disease care model has been organized and after an initial CGA that can be used as the

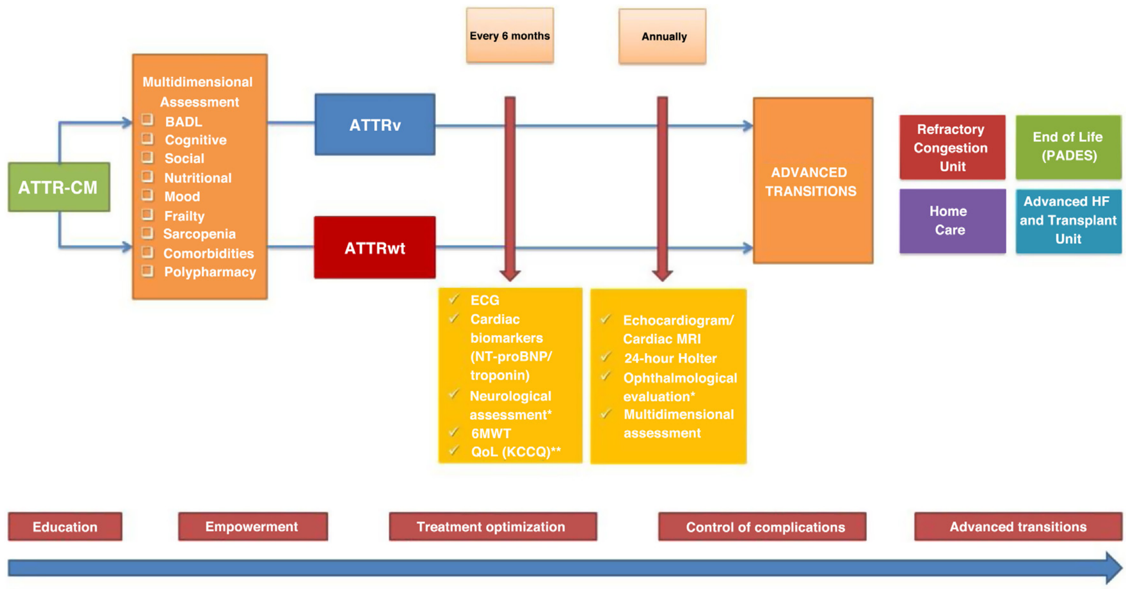



**Figure 6** Global assessment, follow-up, and advanced transitions of patients with ATTR-CM.
6MWT: 6-minute walk test; BADL: basic activities of daily living; ATTR-CM: transthyretin amyloidosis cardiomyopathy; ATTRv: variant transthyretin amyloidosis; ATTRwt: wild-type transthyretin amyloidosis; cardiac MRI: cardiac magnetic resonance imaging; ECG: electrocardiogram; HF: heart failure; KCCQ: Kansas City Cardiomyopathy Questionnaire; NT-proBNP: N-terminal-pro-Brain Natriuretic Peptide; PADES: Home Care and Support Team Program; QoL: quality of life.
*Patients with ATTRv.
**Optional, according to availability and clinical pathway of each healthcare center.

basis for pharmacological and nonpharmacological objectives for follow-up according to current recommendations,[23] a follow-up plan is proposed for these patients with a repeat CGA at least once per year (Figs. 5 and 6).

## Conclusions

It is key to think of CA and its possible manifestations and/or presentations (''red flags'') in order to suspect it. In patients with a high degree of clinical suspicion, it is essential to optimize the diagnostic process as soon as possible in order to improve the prognosis thanks to an early diagnosis. The organization of care based on patient-centered multidisciplinary models is crucial for providing optimal, high-quality care in CA.

## Conflicts of interest

The authors declare that they do not have any conflicts of interest in regard to the content of this work.

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## Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.rceng.2024.04.009.

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